

Course Title	Applied AI (in Python) (AAIP)
ECTS/SWS	3 ECTS / 3 SWS
Course Type	Übung mit Vorlesung (UV)
Lecturer	Dominik Geng, dominik.geng@plus.ac.at
Course Description	Introduction to Python. Functions, classes and exceptions; simple I/O and the most important standard modules. Python IDEs and frameworks for computation (partly cloud-based), special tool-boxes (pandas, matplotlib, numpy, scipy, scikit-learn). Tool-boxes and scripting of these, implementation of classical exploratory data analysis and presentation of the results. Construction of simple Machine Learning models and neural networks from scratch.
Learning Outcomes	Students are able to solve simple problems that they know from other programming languages using the Python language. They can create independent scripts as well as notebooks and know the advantages and disadvantages of both. The students know the various libraries and frameworks for evaluating different data and can use these applications to read data and can use these applications to read clean, process, and display data. They can distinguish between different categories of data and how they can be best visualized. Students are familiar with the background of artificial intelligence and are able program respective models in python.
Evaluation Type	5-point grading scale (Excellent, Good, Satisfactory, Sufficient, Insufficient)

Assessment Methods / Evaluation Criteria

This course is of *immanent character* (UV), which means that all defined criteria must be fulfilled with a positive result. Students are responsible for achieving the minimum requirement of 50 percentage points in total to obtain a passing grade.

Format	Exam modality	Weighting (%)
Checkups	Regular Checkups during class	20%
Lab Assignments	Tick-List and submission	10% ¹
Participation	Oral Questions to Reading+Wrap Up and Presentation of the lab submission	30%
Project	Presentation and Submission	40% ²

Checkups: Short closed-book checkups.

Lab Assignments: at least 50% of the lab assignments have to be completed to get a positive grade.

Participation: Questions to each reading, wrap-up exercises or to lab assignments. This can be an interview/presentation style but can also involve discussion with the group or presentation as a group of students. Not being able to answer the questions leads to a deduction of the points. In case of the lab assignment presentation, also the points for the ticks will not be awarded.

Project: Introduction to the project is given separately.

If a student is unable to *defend* or justify their own work (e.g., items checked off on the list but the student cannot answer related questions), the presentation will be graded with 0% and the entire checklist for the assignment will also be scored with 0%.

Grading Scale

5-point Grading scale	1	Excellent	$\geq 93.0\%$
	2	Good	$\geq 80.0\%$
	3	Satisfactory	$\geq 65.0\%$
	4	Sufficient	$\geq 50.0\%$
	5	Insufficient	$< 50.0\%$

Attendance Requirements, Compensations and Consequences

Attendance in class is mandatory. Students **may miss at most 3** classes. No compensation work is permitted if the minimum attendance requirement is not met.

Owing to the immanent character (UV) of this course, a failing grade requires a full course retake in the following academic year. Individual components cannot be repeated separately.

¹With missing attendance no ticks can be accounted.

²Attendance for final presentation mandatory for project to be graded.

All submission deadlines are strict; extensions will not be granted. In case extensions are granted due to unforeseen situations these will be valid for all and not for individual students only. Such a case will be communicated via e-mail/lms.

Recommended Reading / Learning Resources

François Chollet and Matthew Watson. *Deep Learning with Python*. Manning, third edition, 2025. ISBN 9781633436589. URL <https://deeplearningwithpython.io/>.

Mark Lutz. *Learning Python*. O'Reilly Media, Sebastopol, CA, sixth edition, February 2025. ISBN 9781098171308. URL <https://learning-python.com/about-lp6e.html>.

Simon J.D. Prince. *Understanding Deep Learning*. The MIT Press, 2023. URL <http://udlbook.com>.

Python Software Foundation. The Python Tutorial, 2026. URL <https://docs.python.org/3/tutorial/index.html>.